

Bank Clearing Workspace — Exploration Notes

Field	Value
Version	1.1
Date	June 2026
Type	Historical — pre-restructure as-is analysis (June 2026)
Current architecture	Implementation plan §3
Implementation follow-up	bank-clearing-workspace-implementation-plan.md
Related redesign	admin-portal-redesign-plan.md §7.6

This document captures how the **Bank clearing** workspace worked **before** the June 2026 restructure. It informed the implementation plan. For the **current** system, see [implementation plan §3](#) and [Appendix C smoke checklist](#).

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1. Architecture overview

The workspace is a **Filament resource** (`BankAccountsResource`) whose Eloquent model is `BankStatement`, but the list page (`ListBankAccounts`) is a **multi-tab workspace** over three concepts:

- `BankStatement` — import batches
- `BankTransaction` — CSV lines and synthetic operational rows
- `Transaction` — master **bank** account ledger rows

It uses a **custom content() schema** (not the default Filament list layout): channel switcher → optional Filament tabs → optional imports sub-pills → embedded table.

flowchart TB

```
subgraph nav [Sidebar]
    BC["Bank clearing<br/>badge: uncleared count"]
```

```

end

subgraph page [ListBankAccounts]
  INS[BankAccountsInsightsWidget]
  W[bank-workspace.blade.php<br/>Channel: Bank | SMS]
  subgraph bankChannel [channel=bank]
    FT[Filament tabs x4]
    IP[imports-section-pills<br/>Unmatched | Matched]
    T[EmbeddedTable]
  end
  end
  subgraph smsChannel [channel=sms]
    ST[SMS sub-pills]
    SW[SmsTransactionsTableWidget<br/>or SmsImportSessionsTableWidget]
  end
  end
end

nav --> page
W --> bankChannel
W --> smsChannel
INS --> page

```

List page content order (bank channel):

1. BankAccountsInsightsWidget (header widget, full width)
2. bank-workspace.blade.php — Bank / SMS channel pills
3. Filament tab bar (getTabsContentComponent())
4. imports-section-pills.blade.php — only when tab = imports
5. Embedded data table

2. Entry points & file map

2.1 Primary UI

Role	Path
Resource	app/Filament/Tenant/Resources/BankAccounts/BankAccountsRes
List page	app/Filament/Tenant/Resources/BankAccounts/Pages/ListBankA
Statement detail	app/Filament/Tenant/Resources/BankAccounts/Pages/ViewBankS
Workspace shell	resources/views/filament/tenant/resources/bank-accounts/p
Imports sub-pills	resources/views/filament/tenant/resources/bank-accounts/p

2.2 Tables

Class	Path
Pending operational clearance	app/Filament/Tenant/Resources/BankAccounts/Tables/Pend
Statement lines (CSV)	app/Filament/Tenant/Resources/BankAccounts/Tables/Bank
Master bank ledger	app/Filament/Tenant/Resources/BankAccounts/Tables/Mast
Statement batches	app/Filament/Tenant/Resources/BankAccounts/Tables/Bank

2.3 Actions & services

Role	Path
Row/bulk bank actions	app/Filament/Support/BankTransactionTableActions.php
Import header actions	app/Filament/Support/BankWorkspaceImportTableHeaderActions.php
Match / scope logic	app/Services/BankClearingMatchService.php
Insights data	app/Services/BankAccountsInsightsService.php
Insights widget	app/Filament/Tenant/Widgets/BankAccountsInsightsWidget.php
Insights blade	resources/views/filament/tenant/widgets/bank-accounts-insights.php
Synthetic statement buckets	app/Support/BankStatementBuckets.php
Pending line deletion	app/Services/PendingOperationalClearanceDeletionService.php
Import CSV	app/Services/BankImportService.php
Mirror to cash	app/Services/FundFlowService.php

2.4 Master account relation managers

RM	Path
Pending bank match	app/Filament/Tenant/Resources/MasterAccounts/RelationManagers/PendingBankMatch.php
Bank lines awaiting posting	app/Filament/Tenant/Resources/MasterAccounts/RelationManagers/BankLinesAwaitingPosting.php
Transaction history	app/Filament/Tenant/Resources/MasterAccounts/RelationManagers/TransactionHistory.php

2.5 SMS widgets (embedded in workspace)

Widget	Path
SMS transactions	app/Filament/Tenant/Widgets/SmsTransactionsTableWidget.php
SMS import sessions	app/Filament/Tenant/Widgets/SmsImportSessionsTableWidget.php

3. Navigation layers

Three independent dimensions on the bank channel:

Layer	State property	URL param	Default
Channel	<code>\$channel</code>	<code>?channel=bank\ sms</code>	<code>bank</code>
Main tab	<code>\$activeTab</code> (Filament)	<code>?tab=</code>	<code>clearance</code>
Imports section	<code>\$importsSection</code>	<code>?importsSection=unmatched\ matched</code>	<code>unmatched</code>

SMS channel adds a fourth:

Layer	State	URL	Default
SMS sub-tab	<code>\$smsSubTab</code>	<code>?smsSubTab=transactions\ history</code>	<code>transactions</code>

3.1 Tab resolution (`BankAccountsResource::resolveListBankAccountsTab()`)

Priority:

1. If `resolveChannel() === 'sms'` → return 'sms' (no bank table).
2. Else if Livewire `ListBankAccounts` has `activeTab` → use it.

3. Else `request('tab')` or `'clearance'`.

Aliases:

Incoming <code>?tab=</code>	Resolved
<code>transactions</code>	<code>imports</code>
<code>ledger, statements, imports, clearance, sms</code>	<code>unchanged</code>
<code>anything else</code>	<code>clearance</code>

3.2 Channel resolution (`resolveChannel()`)

1. Livewire `$channel` on `ListBankAccounts` if valid.
2. Else `request('channel')` or `'bank'`.

3.3 URL builder (`BankAccountsResource::listUrl()`)

- Omits `tab` when `$tab === 'imports'` (legacy default for statement lines).
- Omits `channel` when `bank`.
- Supports `filters, smsSubTab=history`.

Implication: Default index URL opens **Pending bank match** (`clearance`), not statement lines.

3.4 Sidebar navigation

Attribute	Value
Label	Bank clearing
Group	Accounts (<code>TenantNavigation::GROUP_ACCOUNTS</code>)
Sort	<code>SORT_BANK_CLEARING = 10</code>
Slug	<code>/bank-accounts</code>
Badge	<code>BankTransaction::uncleared()->count()</code> (warning)
Model label (internal)	Bank statement
Plural model label	Bank accounts

Badge mismatch: Sidebar counts **all** uncleared `BankTransaction` rows. The **Pending bank match** tab badge uses `BankClearingMatchService::pendingOperationalClearanceCount()` only (synthetic operational rows).

4. Bank channel tabs

Defined in `ListBankAccounts::getTabs()`:

Key	Label	Icon	Badge	Table
<code>clearance</code>	Pending bank match	link	pending operational count	<code>PendingOperationalClearanceTable</code>
<code>imports</code>	Statement lines	queue-list	—	<code>BankTransactionsTable</code>
<code>ledger</code>	Master bank ledger	book-open	—	<code>MasterBankLedgerTable</code>
<code>statements</code>	Statements	document-text	—	<code>BankStatementsTable</code>

Default active tab: `clearance` (`getDefaultActiveTab()`).

Tab styling: `data-ff-tab-key + data-ff-tab-color` via `TabLabelColors::forKey()`.

4.1 Table query (`ListBankAccounts::getTableQuery()`)

Tab	Query
ledger	Transaction where account_id = master bank
imports	applyRealBankStatementLinesScope() + status filter from importsSection
clearance	applyPendingOperationalClearanceScope()
default (statements)	BankStatement Eloquent query from resource

Imports section filter:

importsSection	status IN
unmatched (default)	imported, mirrored
matched	posted, duplicate, ignored

4.2 Imports sub-pills UI

Shown only when `resolveListBankAccountsTab() === 'imports'`.

- Amber banner: **Statement lines — action required first**
- Pills: **Unmatched** | **Matched** / **closed**
- Helper copy differs per section (post/clear vs read-only audit)

5. Table configurations

5.1 PendingOperationalClearanceTable

Rows: Uncleared synthetic operational BankTransaction rows (deposits, cash-outs, expense/fee/invest disbursements, invest returns).

Columns:

Column	Notes
transaction_date	sortable
clearance_kind	Virtual — state() from FK ids; badge; searchable(false) required
member.name	sortable
amount	money, color by sign
description	searchable
status	badge

Type badge mapping (clearance_kind state):

Condition	Label	Color
invest_return_id	Return in	success
invest_disbursement_id	Invest out	warning
fee_disbursement_id	Fee	info
expense_disbursement_id	Expense	danger
cash_out_request_id	Cash out	warning

Condition	Label	Color
default	Deposit	success

Row actions: View (`ViewBankTransactionAction`), **Clear / Match**, **Remove pending bank match**

Bulk: Clear / match, bulk remove pending matches

Empty state: Accepted deposits, cash-outs, expense disbursements... still need a matching line from an imported bank statement.

5.2 BankTransactionsTable

Rows: Real CSV statement lines (excludes synthetic operational buckets).

Columns: date, source (statement filename), amount, assigned member, master cash summary, status, cleared icon, duplicate-of (hidden)

Row actions: View, **Post to cash**, **Post to member**, **Clear / Match** (pending operational rows only), **Ignore**, **Delete**

Bulk: Clear / match, post to cash, post to member, ignore, delete

Filters: status, cleared, date, member, duplicate link, statement, description contains

Empty state: Imported bank statement lines. Manual master bank credits/debits are on the Master bank ledger tab.

5.3 MasterBankLedgerTable

Rows: Transaction on master **bank** account (not `BankTransaction`).

Header actions: Credit / Debit / Refund (`AccountTransactionManualAdjustmentHeaderActions`)

Row actions: View only (`ViewAccountTransactionAction`)

5.4 BankStatementsTable

Rows: Import batch metadata.

Row click: navigates to `ViewBankStatement`

Actions: View modal, delete (cascades transactions)

Bulk: delete selected, refresh

5.5 ViewBankStatement page

- Detail insights + statement metadata schema
- `BankTransactionsRelationManager` — per-statement lines (view + delete)

6. Domain scopes & services

6.1 BankClearingMatchService scopes

Method	Purpose
<code>applyRealBankStatementLinesScope()</code>	Excludes synthetic operational statement filenames
<code>applyPendingOperationalClearanceScope()</code>	Uncleared + operational FK + operational bucket filenames

Method	Purpose
<code>applyPendingOperationalClearanceScopeForMasterAccountType()</code>	Filter by master account type
<code>applyBankLinesAwaitingPostingScope()</code>	Real CSV lines with status <code>imported</code> or <code>mirrored</code>

6.2 Operational clearance scope (SQL shape)

```
bank_transactions WHERE
  is_cleared = 0
  AND (fund_posting_id OR cash_out_request_id OR expense_disbursement_id
        OR fee_disbursement_id OR invest_disbursement_id OR invest_return_id IS NOT NULL)
  AND bank_statement.filename IN (operational clearance buckets)
```

6.3 Master account types with pending clearance

cash, expense, fees, invest — via `masterAccountTypesWithPendingClearance()`.

6.4 Count helpers

Method	Counts
<code>pendingOperationalClearanceCount()</code>	All pending operational rows
<code>pendingOperationalClearanceCountForMasterAccountType()</code>	Sub-account count type
<code>bankLinesAwaitingPostingCount()</code>	CSV lines needing post to cash pool

6.5 Synthetic vs real buckets

Defined in `App\Support\BankStatementBuckets`:

- **Operational clearance** — member-postings, member-cash-outs, master-expense-disbursements, etc.
- **Synthetic operational** — excluded from “real bank statement lines” scope

7. User workflows

A. Import bank CSV

1. Bank clearing → Bank channel.
2. **Import statement** (page header on clearance, imports, statements tabs — not ledger).
3. Modal: CSV, bank template, optional bank name → `BankImportService::importCsv()`.
4. New rows → **Statement lines** → **Unmatched** (`status=imported`).

B. Post import line to master cash

1. Statement lines → Unmatched → **Post to cash** (or bulk).
2. `FundFlowService::mirrorToCash()` → status `mirrored`, master cash credited.

C. Assign import line to member

1. **Post to member** on `imported` or `mirrored` row.
2. Credits/debits member cash via posting services.

D. Operational clearance (deposit, cash-out, expense, etc.)

1. Business flow creates synthetic `BankTransaction` on operational statement bucket.
2. Row appears on **Pending bank match** (default tab).
3. **Clear / Match** → select real CSV line → `BankClearingMatchService::clearMatchPair()`.
4. Row leaves pending list; import line marked cleared.

E. Match from statement lines

Same **Clear / Match** UI on `BankTransactionsTable` for eligible rows — duplicate entry point.

F. Ignore / delete

- **Ignore:** imported status only.
- **Delete:** `BankTransactionDeletion` rules; pending operational delete uses `PendingOperationalClearanceDeletionService` (ledger reversal).

G. Manual master bank adjustments

Master bank ledger tab → Credit / Debit / Refund.

H. Review import batches

Statements tab → row → `ViewBankStatement`.

I. SMS channel

1. Channel pill → **SMS**.
2. **Transactions:** `SmsTransactionsTableWidget` — review/post to member cash.
3. **History:** `SmsImportSessionsTableWidget` — **Import SMS file** in widget header.

J. From master account pages

- **Pending bank match** RM reuses `PendingOperationalClearanceTable` (scoped).
- **Bank lines awaiting posting** RM on master cash — post actions mirror statement lines.
- **Transactions** RM — ledger with bank import column on cash accounts.

8. Header actions & widgets

8.1 Page header actions (`ListBankAccounts::getHeaderActions()`)

Tab	Import statement
imports, transactions, clearance, statements	Yes
ledger	No
SMS channel	No

Uses `BankWorkspaceImportTableHeaderActions::bankStatementImportAction()`.

8.2 `BankAccountsInsightsWidget`

Always mounted as header widget (`getHeaderWidgets()`).

On clearance / imports tabs — clearing KPI strip (4 cards):

- Imported today, Auto-matched, Unmatched, Stale pending (links via `BankAccountsInsightsService`)

Always shown:

- Hero banner + KPI strip
- Posting pipeline status bars
- Master cash / master bank balance cards
- Import trend, recent statements

Widget view: `ff-app-insights` CSS class.

Data driven by `BankAccountsInsightsService::snapshot()` which reads `resolveListBankAccountsTab()` for context.

9. Master account overlap

On `ViewMasterAccount (MasterAccountResource)`:

Relation manager	Visible when	Reuses
Bank lines awaiting posting	master cash	Custom inline table + post actions
Pending bank match	master cash , expense , fees , invest	<code>PendingOperationalClearanceTabl</code>
Transactions	all	Standard ledger

Pending bank match RM nuance:

`showClearanceKindColumn: in_array($owner->type, ['cash', 'invest'], true)`

Expense and fees master accounts **hide** the Type column but still show the pending match table.

Duplication: Same Clear/Match and Post workflows exist on workspace tabs and on master account RMs — useful as contextual drill-down but scatters muscle memory.

10. SMS channel

Rendered entirely inside `bank-workspace.blade.php` when `$channel === 'sms'`:

- No Filament tabs
- No embedded `BankAccountsResource` table
- Livewire widgets replace the table
- Import entry: **Import SMS file** on History widget (not page header)

Copy:

- Workspace helper: *Switch between bank statement and SMS channels. Parsing templates are managed under Settings.*
- Transactions: *Review parsed SMS transactions and post verified rows to member cash.*
- History: *Monitor SMS import batches with counts, errors, and completion state.*

11. Cross-links from reconciliation

Source	Target
<code>ReconciliationOverviewPage::getBankClearingUrl</code>	<code>BankAccountsResource::listUrl('clearance')</code>
<code>reconciliation-workspace-shortcuts.blade.php</code>	Bank clearing shortcut card
<code>ReconciliationExceptionPresenter</code>	<code>domainLabel('bank_clearing')</code> , context URLs to <code>listUrl('clearance')</code>

Reconciliation treats **Pending bank match** as the primary bank clearing entry.

12. UI patterns & CSS

Class / attr	Usage
<code>ff-tenant-tab-pills, __item, __item--active</code>	Channel, SMS, imports-section pills
<code>ff-app-insights</code>	Bank accounts insights widget
<code>data-ff-tab-key, data-ff-tab-color</code>	Filament bank tabs
<code>ff-recon-shortcut-card</code>	Reconciliation → bank clearing links

Filament table standards: filters, grouping, bulk actions enforced via `FilamentTableStandardsTest`.

`Global Column::configureUsing()` applies `->searchable()` to all columns — **virtual columns must opt out** with `->searchable(false)`.

13. Copy & labels inventory

Location	English
Page / nav title	Bank clearing
Channel	Bank / SMS
Tabs	Pending bank match · Statement lines · Master bank ledger · Statements
Imports sub	Unmatched · Matched / closed
SMS sub	Transactions · History
Primary import CTA	Import statement / Import SMS file
Match action	Clear / Match
Post actions	Post to cash · Post to member
Delete pending	Remove pending bank match
Master RM titles	Bank lines awaiting posting · Pending bank match

Arabic: `tests/Feature/Tenant/BankAccountsTranslationTest.php`.

14. Pain points

Ranked by operational impact:

1. **Three navigation layers** on bank channel (channel → tab → imports sub-pill).
 2. **Default tab is clearance** while many sessions start with unmatched imports; redesign plan §7.6 envisioned unmatched-first.
 3. **Duplicate Clear/Match** in Pending tab, Statement lines tab, and master RMs.
 4. **Duplicate posting** — Bank lines awaiting posting RM vs Statement lines.
 5. **Overlapping vocabulary** — uncleared, pending match, unmatched, imported/mirrored/posted, cleared.
 6. **Heavy insights** above queue competes with table on small viewports.
 7. **Sidebar badge tab badge** — different counting logic.
 8. **Legacy naming** — slug `bank-accounts`, plural label “Bank accounts”, nav “Bank clearing”.
 9. **Hybrid tab systems** — Filament tabs + custom pills (unlike Reconciliation’s single registry).
 10. **SMS cohabitation** — different import entry points and widget embedding.
 11. **Import on clearance tab** — header action on tabs that are not strictly about importing.
-

15. Technical hazards

15.1 Virtual column global search

`TextColumn::make('clearance_kind')` uses `state()` only — no DB column. Global `->searchable()` caused:

Unknown column 'clearance_kind' in 'WHERE'

Fix applied: `->searchable(false)` on that column.

Audit needed: Other `->state()` columns across Filament tables (member balances, loan outstanding, fund tier computed columns, etc.).

15.2 Tab reconfiguration on switch

`ListBankAccounts::updateActiveTab()` calls `reconfigureTableForActiveTab()` — rebuilds table, resets filters. Switching tabs loses filter state (by design).

15.3 `listUrl()` default tab omission

`listUrl()` omits `tab` param when `$tab === 'imports'`, but default resolved tab is `clearance` — deep links must pass `tab` explicitly for non-imports views.

16. Test coverage

Test file	Covers
<code>BankAccountsListTabResolutionTest.php</code>	Tab/channel URL resolution
<code>BankAccountsImportHeaderActionTest.php</code>	Import action visibility per tab
<code>SmsImportSessionsAndBankWorkspaceTest.php</code>	SMS channel, widgets
<code>BankClearingMatchServiceTest.php</code>	Scopes, matching
<code>MasterAccountPendingBankPostingScopeTest.php</code>	Per-account pending scopes
<code>PendingOperationalClearanceDeletionTest.php</code>	Delete pending + table search regression
<code>BankAccountsTranslationTest.php</code>	Arabic strings
<code>AdminPortalPhaseEightTest.php</code>	Tab defaults (clearance)
<code>ReconciliationExceptionPresenterTest.php</code>	Bank clearing context links

17. Reconciliation patterns to mirror

The reconciliation workspace redesign provides a proven template:

Pattern	Reconciliation	Bank clearing today
Tab registry	ReconciliationTabRegistry	Keys in <code>getTabs()</code> + <code>resolveListBankAccountsTab()</code>
Custom shell blade	<code>reconciliation.blade.php</code>	<code>bank-workspace.blade.php</code> (channel only)
Pill navigation	<code>reconciliation-tab-pills</code>	Mixed Filament tabs + pills
Workspace actions	<code>workspacePanelActions()</code> + partial	Page <code>getHeaderActions()</code> only
Overview KPIs	Queue insights partial	Full insights widget always
Shortcut cards	<code>reconciliation-workspace-shortcuts</code>	Inbound only (from recon)
Presenter	ReconciliationExceptionPresenter	None for bank queue rows
Grouped actions	ReconciliationExceptionActions	Flat row action lists
Embed trait	EmbedsAsAuditWorkspacePanel	Not used

18. Post-implementation reference

Added June 2026 after Phases 1–4 shipped. Supersedes sections 1–17 for day-to-day navigation.

18.1 Architecture (current)

flowchart TB

```
subgraph nav [Sidebar Finance]
    BC["Bank clearing<br/>badge: openCount()"]
    SMS["SMS imports"]
end

subgraph bankPage [ListBankAccounts /bank-accounts]
    SHELL[bank-clearing.blade.php]
    subgraph tabs [Tabs]
        Q[Work queue + BankClearingQueueTable]
        L[Bank ledger + MasterBankLedgerTable]
        H[Import history widgets]
    end
end

subgraph smsPage [SmsImportsPage /sms-imports]
    ST[SmsTransactionsTableWidget]
    SH[SmsImportSessionsTableWidget]
end

nav --> bankPage
nav --> smsPage
SHELL --> tabs
H --> Batches[BankImportBatchesTableWidget]
H --> Closed[BankClosedStatementLinesTableWidget]
```

18.2 Entry points (current)

Role	Path
Bank clearing list	app/Filament/Tenant/Resources/BankAccounts/Pages/ListBankAccounts.php
Workspace shell	resources/views/filament/tenant/pages/bank-clearing.blade.php
SMS imports page	app/Filament/Tenant/Pages/SmsImportsPage.php
Unified queue table	app/Filament/Tenant/Resources/BankAccounts/Tables/BankClearingQueueTable.php
Queue actions	app/Filament/Support/BankClearingQueueActions.php
Tab registry	app/Filament/Tenant/Support/BankClearingTabRegistry.php

Removed: bank-workspace.blade.php, imports-section-pills.blade.php, Bank/SMS channel on bank clearing page.

18.3 User-facing vocabulary (current)

Label	Meaning
Match to bank line	Pair operational/import row with CSV evidence
Clear without evidence	Close operational synthetic row without CSV line
Match automatically	Unique candidate within tolerance
Open in bank clearing	Deep link from master account RM with <code>queueFilter</code>

18.4 Pain points — resolution status

#	Pre-restructure pain	Resolution
1	Three navigation layers	Single tab row + queue filter chips
2	Wrong default tab	Default Work queue
3	Duplicate Clear/Match	Split actions; RMs read-only + link
4	Duplicate posting RM	View-only + link to queue
5	Overlapping vocabulary	Clear vs match labels
6	Heavy insights on queue	Slim KPIs + optional toggle
7	Badge mismatch	<code>openCount()</code>
8	SMS cohabitation	Separate <code>/sms-imports</code> page

18.5 Test coverage (current)

Test file	Covers
BankClearingWorkspaceLocaleSmokeTest.php	EN/AR shell, queue, history, SMS page
BankClearingWorkspaceSmokeTest.php	Tab switching, URLs, import action
BankClearingQueueWorkflowTest.php	Post, match, clear-without-evidence, combined history
BankClearingPhaseThreeAndSmsTest.php	RM preview, SMS redirect
BankClearingTabRegistryTest.php	Tab aliases
BankClearingMatchServiceTest.php	Matching, auto-match, clear without evidence

Appendix — ListBankAccounts Livewire URL state (historical)

Property	#[Url] name	Notes
\$channel	channel	Removed; legacy redirects to SmsImportsPage
\$smsSubTab	smsSubTab	Moved to SmsImportsPage::\$activeTab
\$queueFilter	queueFilter	Active
\$historySection	historySection	Expands closed-lines panel when closed
\$activeTab	(Filament)	queue ledger history

Historical exploration. For current behaviour see [bank-clearing-workspace-implementation-plan.md](#).